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Personality Development and Concepts of Structure

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Source: *Child Development*, Sep., 1968, Vol. 39, No. 3 (Sep., 1968), pp. 671-690

Published by: Wiley on behalf of the Society for Research in Child Development

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PERSONALITY DEVELOPMENT AND CONCEPTS OF STRUCTURE

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It is proposed that many concepts of personality structure and structural development are of primary relevance to one of three general orientations, here called the "classical," "differential," and "ipsative" views. Certain theoretical and methodological features of each approach are reviewed. The discussion emphasizes differences and similarities among the views, and points out some implications for research.

A number of recent studies of personality development have considered the question of structural consistency and change. While much of this work has been inspired by theoretical considerations, it is often difficult to determine whether a particular operational concept of structure is of greater relevance to one theory than to another. One reason for this interpretive gap is that traditional theories of personality development typically are couched in broad terms that do not specify translations required at the operational level. Nevertheless, it is possible to discern theoretical lines of similarity and dissimilarity among concepts of structural development found in the literature. The present thesis is that such concepts are of primary relevance to one of three views. Stated in brief, these are: *classical* developmental theory, positing an invariant sequence of personality stages; *differential* analysis, which considers how in the course of development individuals become "sorted" into subgroups differentiated according to status and behavioral attributes; and the *ipsative* approach, which looks at intraindividual consistencies and change in the organization of attributes over time. These three views will be examined in some detail, with emphasis upon differences

This paper was written while the author was U. S. Public Health Service visiting scholar, Personality Research Group, Educational Testing Service. The work was supported in part by the National Institute of Child Health and Human Development, under research grant 1 PO1 HDO1762-01. Author's address: Human Development Research Group, Educational Testing Service, Princeton, New Jersey 08540.

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in how each characteristically deals with certain problems of conceptualization and methodology.

The discussion will be limited in several respects. The primary concern is with *structural* development, that is, with developmental consistency and change in the organization of personality attributes. Little attention is given to other issues in developmental personality theory, such as the nature of the environmental determinants of attribute acquisitions. Because they raise distinct questions requiring separate treatment, the concepts of "fixation" and "regression" are mentioned only in passing. Finally, the intent is to present only the main outlines of the three structural views; not considered are many possible refinements of conceptualization, measurement, and analysis within each approach.

THE CLASSICAL DEVELOPMENTAL VIEW

Concept of Structure

The main prototypes of classical developmental analysis are psychoanalytic developmental theory (A. Freud, 1965; S. Freud, 1905; Rapaport, 1960) and Piaget's theory of cognitive development (Flavell, 1963; Kessen & Kuhlman, 1962; Tanner & Inhelder, 1956: 1960). Although they differ in important respects (Wolff, 1960), these theories share the view that individuals pass through a series of qualitatively different levels of structural organization and that the sequence of stage progression is invariant. The present review considers certain recent translations of these assumptions into suitable terms for personality research.

Discussions of personality stage progressions have considered ego development (Loevinger, 1966; Van den Daele, in press), the development of interpersonal relations (Sullivan, Grant, & Grant, 1957), and the development of moral judgment (Kohlberg, 1964). Although it has not yet proved possible to arrive at a concise definition of each stage, Loevinger (1966) suggests that the several accounts refer to a common underlying scale of ego development, consisting of the following seven levels: presocial-symbiotic, impulse ridden, opportunistic, conformist, conscientious, autonomous, and integrated. It is important to note at the outset that while this approach posits the broad applicability of the developmental scale, it does not imply universality in the sense that progression *will* occur, but rather that if progression *does* occur, its course will be predictable and invariant. Consequently, investigators in this tradition are particularly interested in both the rate and extent of developmental progressions (Kohlberg, 1964).

Attribute Measurement

Relevant attributes may change either monotonically or nonmonotonically as a function of age (Harvey, Hunt, & Schroder, 1961; Loevinger,

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1966), and are meaningful for this approach to the extent that they are part of a coherent response pattern that defines a point on the developmental scale. This means that while theory specifies a certain ordering of points on this scale, measurement calls for nominal classifications in accordance with complex decision rules. For example, when determining the child's level of moral development, investigators have assessed the *principle* underlying the child's rationale for moral decisions rather than scaling direction, intensity, or frequency of the contents of the decisions themselves (Kohlberg, 1964; Piaget, 1932).

Structural Analysis within Age Periods

An individual's or group's location on the developmental scale can be determined from the central tendency of a set of responses. This situation is complicated, however, by the distinct possibility that an individual may be at different levels, depending upon the area of personality functioning under consideration, analogous to Piaget's concept of "horizontal décalage" (Flavell, 1963). Thus, for example, since social role performances vary according to the complexity of the interlocking actions called for, individuals probably attain higher levels earlier in the case of the more simply organized role performances. It should be noted that variations among areas of functioning in rate and extent of progression would be consistent with the classical view if invariant sequences were found within each area. It may be that as the child becomes actively engaged in more varied and complex personality tasks in the course of development, his initial handling of each new problem is characterized by lower levels of functioning, with progression occurring over time, perhaps at a rate and to an extent that are area-specific. If this were generally the case, then it would be possible and perhaps useful to define structural "complexity" as the amount of unevenness (interarea variance) within the individual (or group) at a given age period. On the other hand, there may be considerable transfer across areas, perhaps increasing with age, in which case new areas would be rapidly assimilated to the individual's current level of functioning. In any case, research is needed to clarify this issue, perhaps by means of factor analytic procedures that isolate areas having common rates of progression during each age period.

Structural Development

Structural change is defined as progression to the next level or to certain transitional phases (Ausubel, 1958; Harvey, Hunt, & Schroder, 1961). Such shifts are indexed by changes in central tendency, distribution shape, and response variability on the developmental scale, and by the extent to which change generalizes to other personality areas.

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At least three hierarchical models of structural change are consistent with the classical view. In the most radical case, attainment of a higher level fundamentally reorganizes and "displaces" lower levels, which thereby become inoperative and difficult to reactivate. In the second case, higher levels subsume earlier ones, but the latter remain active and quite capable of taking on a dominant role in normal functioning under certain circumstances. The third form occurs when the highest level displaces lower levels and the latter remain latent and re-emerge if special circumstances produce regression. Personality research has not yet clarified which of these alternatives is most applicable.

Since the classical view is essentially a theory of change, it would define structural consistency over time as the *absence* of progression (or regression). In this sense, structural consistency could be due to (*a*) absence of the conditions necessary to produce progression, (*b*) attainment of the highest level, or (*c*) fixation.

Person Sampling

According to fixed sequence theory, every individual who progresses will do so one step at a time and in an invariant order. This hypothesis is most directly tested by longitudinal investigations in which "natural" or experimentally induced level progressions are observed in broadly selected samples. However, the sequential hypothesis can also be tested by comparing the profiles of levels attained by cross-sectional samples of differing age, and by showing that individual differences in level *within* age periods are reduced significantly when factors presumed or known to influence rate and extent of progression are partialed out.

Both intrinsic (e.g., intellectual) and experiential (e.g., cultural) factors are believed to influence stage progressions (Kohlberg, 1964). To test hypotheses concerning rate and extent of progression, this approach utilizes a comparative analysis of developmental curves in subpopulations classified according to these presumed determinants. Analysis proceeds by comparing subgroup trends with respect to the highest level reached, the age at which each group reaches an asymptote for a given level, and differential timing in the appearance of similarly shaped curve segments (Kohlberg, 1966).

Concept of Individuality

Since this approach locates the individual's responses on the developmental scale, individuality may be described in relation to age norms of level attainment. Insofar as differential rates of progression exist among areas of personality functioning and are essentially the same for all subpopulations, it is meaningful to sum an individual's scores across areas to arrive at an estimate of his overall level of attainment. However, even

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assuming sequential invariance within personality areas and for all individuals, there still could be individual × area interactions in *rates* of progression. Since in the latter case an additive procedure would tend to mask the true nature of individual differences, a profile description would be more appropriate.

THE DIFFERENTIAL APPROACH

Concept of Structure

Differential psychology is concerned with discovering dimensions of individual and group differences. "Structures" are defined by the ways that individuals are grouped together and differentiated from other individuals or groups according to status attributes (e.g., sex) and locations on behavioral dimensions (e.g., extraversion-introversion). Broadly speaking, structural *development* refers to the genealogy of the network of differentiated subgroups within the total population. Unlike the classical view, this approach traditionally has emphasized structural consistency rather than change, posits no universal developmental scale, and generally has been unconcerned with sequential arrangements in personality development. Indeed, some investigators have attempted to demonstrate that the dimensions of personality remain essentially invariant over time (Cattell, 1957; Cattell & Coan, 1957; Hundleby, Pawlik, & Cattell, 1965; Peterson & Cattell, change insofar as it attends to age variations in the dimensionality of 1959). However, differential analysis *does* concern itself with developmental personality and considers the stabilities of individual and group differences over time.

Attribute Measurement

The concept of "attribute" in this approach is the familiar one of a trait that orders individual differences. If the factorial structure of a domain is well known, attributes can also be defined in terms of factorial components. This kind of relation between factors and attributes differs from that between levels and their signifiers in the classical approach, for in the latter case, level criteria define specific attribute patterns that lead (ideally) to classifications at one and only one level or transitional phase, at least within each area of personality functioning.

Structural Analysis within Age Periods

The differential approach isolates the basic ways that persons and groups differ by means of correlational, factor, or some form of cluster analysis. Structural *complexity* is signified both by the number of dimensions required to account for most of the common variance and by the

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correlations among dimensions. (If dimensions are correlated, it is possible to isolate higher-order, orthogonal factors.) In recent years, a variety of structural models have been proposed, including two-factor (Peterson, 1960), many-factor (Cattell & Coan, 1957; Digman, 1963, 1965; Stott, 1962), and circumplex (Becker & Krug, 1964; Schaefer, 1961) solutions. These and other analytic procedures have done much to sharpen our understanding of differential structures, but they have not led to a sufficiently comprehensive account of child personality. Perhaps more elaborate models are required, such as the hierarchical approach advocated by certain factor theorists (Burt, 1949; Coan, 1964; Eysenck, 1951; Guilford, 1959; Moursy, 1952). Moreover, the outcome of any differential structural analysis is very much influenced by the heterogeneity of settings, persons, and attributes included, and most analyses to date have been quite limited in these respects. In addition, any general model will need to account for evidence that the structure of personality changes as a function of age, a matter considered in greater detail below.

Structural Development

A number of indices of structural development have been used within the differential approach. It is useful here to distinguish between two types of questions (Ausubel, 1958; Emmerich, 1964). The first question is whether the dimensions ordering individual differences in personality are essentially the same ("continuity") or change ("discontinuity") in the course of development. Structural continuity-discontinuity has been signified by the extent of developmental change in: (1) the *meanings* of dimensions, revealed, for example, by changes in factor loading configurations (Emmerich, 1964, 1966a) or dimensional correlates (Bronson, 1966); (2) the *number* of dimensions required to account for common variance (Black, 1965; Emmerich, 1964, 1966a); (3) the *absolute amount* of common variance accounted for by a single underlying dimension (Wapner & Werner, 1957); (4) the *relative amount* of common variance accounted for by each of several dimensions (Black, 1965; Emmerich, 1964, 1966a); and (5) the nature of the *intercorrelations* among dimensions (Black, 1965; Bronson, 1966; Emmerich, 1964).

The second basic question is whether the ranks of individuals (and distances between ranks) are about the same ("stability") or change ("instability") at different periods of development. Amount of stability in this sense is commonly measured by the magnitude of the (positive) correlation coefficient between the same individuals assessed at different periods (Bloom, 1964).

A thorough analysis of structural consistency and change in this approach deals simultaneously with both questions of continuity-discontinuity and stability-instability. By dichotomizing and cross-classifying these two

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dimensions, we may discern four distinct types of structural development (Emmerich, 1964, 1966a). Stability in the context of continuity reflects the purest case of structural *consistency* in development, and has been of particular interest to differential (but not classical) developmental psychologists interested in when (and whether) the main outlines of later personality are established earlier in life. A second form is continuity accompanied by instability of individual differences. In this case, there is partial structural *change* in the sense that *persons* are redistributed along the same underlying dimensions over time. Another type of partial change is discontinuity with stability, signifying a reorganization of the *attributes* that differentiate individuals from one another at each time period, but with maintenance of individual ranks over time on the transformed dimension (Emmerich, 1964, 1966a, 1966b; Kagan & Moss, 1962). The final case is discontinuity with instability, reflecting the most radical form of structural change.

It should be emphasized that continuity-discontinuity and stability-instability are actually continua, and therefore the four kinds of structural development just considered represent extreme types. It is of particular theoretical importance to determine whether discontinuity reflects a fundamental reorganization of attributes, represented, for example, by the emergence of a totally new dimension, whether it is merely a matter of age-specific variations in the indicators of otherwise continuous dimensions or whether structural change falls somewhere between these two extremes (Yarrow, 1964; Yarrow & Yarrow, 1964). The problem here is to detect the *level* at which discontinuity occurs. Hierarchical factor models would appear to be especially well suited for dealing with this problem (Damarin & Cattell, in press).

Stability criteria range in stringency from statistical significance of a stability coefficient to perfect correlations (after corrections for attenuation) in the total "longitudinal network" (Emmerich, 1964) of stability coefficients across multiple time periods. Again, however, it is important to consider patterns that fall between these extremes. For example, a "sleeper effect" (Kagan & Moss, 1962) may be present when an attribute that is stable between t_1 and t_3 also is unstable between t_2 and t_3.

In order to provide complete chartings of the genealogy of structural development, the differential approach is faced with the formidable but important task of discovering which of these several forms of consistency and change apply to the various dimensions of personality during each time span.

Person Sampling

Because of its programmatic intent to discover the essential ways that individuals and groups differ from one another, differential research has

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considered a large variety of subpopulations and personality traits (Tyler, 1965). Indeed, an ideal strategy might be to sample heterogeneous populations (and attributes) longitudinally in order to maximize the universality of resulting structural generalizations. Such a strategy also is well suited to the classical approach, but its essential purpose differs in the two cases. Whereas differential analysis strives toward a comprehensive understanding of the dimensions that partition individuals and groups from one another at each age, the classical developmentalist would use such information to test hypotheses about universality, rate, and extent of progression on the developmental scale.

Concept of Individuality

Individuality is defined in the usual psychometric sense as one's location in multidimensional space. Comprehensive longitudinal investigations make it possible to chart developmental consistencies and change in the individual's location on each dimension.

IPSATIVE ANALYSIS

Concept of Structure

The ipsative concept of structure refers to the intraindividual organization of behavioral attributes. For reasons that have been reviewed elsewhere, personality research typically has not employed an ipsative methodology (G. W. Allport, 1961; Broverman, 1962; Holt, 1962; Mowrer, 1953; Stephenson, 1953), although notable exceptions are found in the work of G. W. Allport (1961), Murray (1938), Rogers and Dymond (1954), and Stephenson (1953). With respect to structural development in particular, there have been only a few attempts to apply this approach in research (Escalona & Heider, 1959; Martin, 1964; Murphy, 1962, 1964; Rabson, 1966; Smock, 1966; Thomas et al., 1963). Consequently, the exposition that follows deals more extensively with preliminary matters of conceptualization.

Attribute Measurement

The ipsative approach traditionally has emphasized the importance of "unique" as well as "common" traits (G. W. Allport, 1961, 1962), although by default or otherwise even the champions of ipsative measurement no longer emphasize this distinction (G. W. Allport, 1966; Block, 1961; Broverman, 1962). Here as well as elsewhere in the discussion it is important to distinguish between traits as structural *elements*, on the one hand, and traits as structural *organizations*, on the other. At the level of elements

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(henceforth referred to as "attributes"), any unique attribute is also common, for the only way that the two might be distinguished is in terms of the nature of the distribution of individuals on the attribute, a matter of some interest but certainly not of sufficient importance to perpetuate the issue. There is also the complicating fact that new personality attributes become differentiated in the course of development, opening up the possibility that such emergents will be unique. But even here it would seem reasonable to expect the total universe of attributes based upon interindividual differences within all age periods to be isomorphic with that for intraindividual differences between all adjacent age periods. Thus, the question of uniqueness would seem to pose no special problem of attribute sampling for the ipsative approach.

A related concern is whether it is legitimate to judge differing substantive attributes on a common scale, for example, whether it is meaningful to say that an individual is more "dependent" than "aggressive." It is now reasonably clear that such a procedure poses no insurmountable problems of scaling and that it can be applied reliably and usefully for understanding personality (G. W. Allport, 1962; Block, 1961; Broverman, 1962). But if the above distinction between elements and structures is taken seriously, then what does become critical is recognition that these two aspects of personality call for distinct assessment procedures. No matter how adequately scaled, ipsative profiles based upon frequency, amount, or intensity carry little or no information concerning the *interrelations* among profile attributes within the person. And it is from these interrelations that the key to understanding ipsative structural organization is to be found (G. W. Allport, 1961, 1962, 1966). This requirement of utilizing distinct measures of attribute repertoires *and* attribute interrelations differs from the classical approach in which attributes are derived from level criteria on the developmental scale. It also contrasts with the differential approach, where the assessment of individual differences on a set of attributes provides the essential data used in structural (dimensional) analysis.

Structural Analysis within Age Periods

The task of ipsative structural analysis thus becomes one of determining the nature of the interrelations among attributes within the individual's repertoire. While this problem has been considered by social (e.g., Festinger, 1957; Heider, 1958), personality (G. W. Allport, 1962), and developmental (Lewin, 1951) theorists, there has not yet evolved a systematic framework that places ipsative personality analysis in a developmental context. The discussion therefore turns to a possible approach to this problem.

Recent psychological theorizing has drawn our attention to the role of hierarchical structures in which higher levels influence, regulate, or control lower levels (Berlyne, 1965; Chomsky, 1957; Miller, Galanter, & Pribram,

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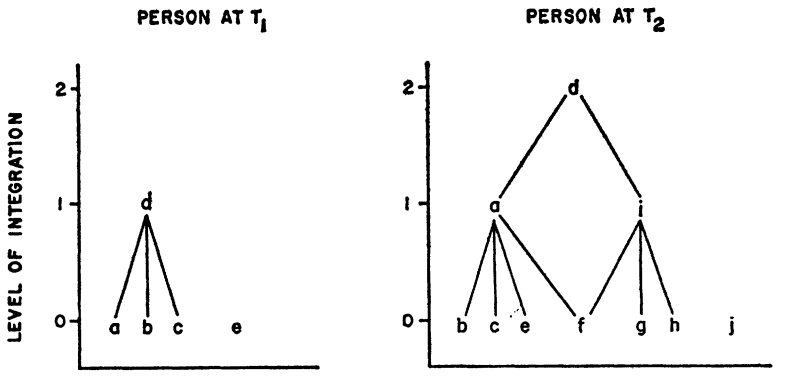
1960; Moursy, 1952; White, 1965). Among developmental theories, perhaps the most explicit statement of the importance and pervasiveness of hierarchical organization is found in Werner's orthogenetic principle (Werner, 1948, 1957). What is surprising, however, is that while this principle clearly refers, among other things, to developmental processes of differentiation and hierarchical integration *within intraindividual structures,* it has not been translated into an ipsative framework. What seems to have happened instead is that classical analysis has been used to test the principle's generality, while differential analysis has been applied to studies of individuality (Werner, 1957; Witkin, 1960). Nevertheless, the orthogenetic principle is especially well suited to an ipsative translation because it distinguishes so clearly between the concepts of differentiation and hierarchical integration, just as the present discussion has insisted that ipsative personality repertoires and their organization must be kept distinct.

Figure 1 illustrates the (early) development of an ipsative personality structure. It represents, of course, only one of many possible outcomes within this framework.

A variety of structural properties can now be identified. From a qualitative point of view, it would be of interest to locate each attribute's level in the hierarchy, especially those high in the structure that serve as "organizers" of elements located at lower levels. Quantitative structural indexes include the number of distinct elements in the individual's repertoire, the number of levels of hierarchical organization, and the number of subhierarchies that serve as organizers within the total structure.

As noted earlier, structures of this kind are derived by means of scaling procedures that reflect the nature of interrelations among attributes within the individual's repertoire. Such procedures are still rather primitive, but



ELEMENTS (a, b, c, etc.) REPRESENT DIFFERENTIATED PERSONALITY ATTRIBUTES

FIG. 1.—Example of development of an ipsative structure

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the literature does provide suggestions as to possible ordering principles (G. W. Allport, 1962). For example, consideration has been given to the relative strengths of associative bonds between attributes, as indexed by the probability of their contiguous expression (G. W. Allport, 1965, 1966; Baldwin, 1942). It is also possible that "serials" (Murray & Kluckhohn, 1953), "teleonomic trends" (F. H. Allport, 1937; F. H. Allport & Frederiksen, 1941), or "plans" (Miller, Galanter, & Pribram, 1960) serve as regulators of subsumed attributes. The common distinction between goals and instrumental acts suggests that it might prove feasible to define higher levels in terms of relatively invariant goal states (within a given period of development), while lower-order attributes become vicarious instrumental dispositions in the service of these goals. In this regard, there may be individual differences in attribute relations (Rabson, 1966), implied, for example, when the inference is made that one person is dependent "because" he is more basically aggressive, whereas another is judged as aggressive "because" he is basically dependent. In any case, it remains an important part of the unfinished business of ipsative structural analysis to formalize and scale potential ordering principles such as these.

Structural Development

Developmental analysis deals with consistencies and change in both the qualitative and quantitative features of ipsative structures. Such analyses probably would reveal certain laws paralleling those already well established by other approaches. For example, we might expect the number of differentiated attributes to increase with age throughout childhood and adolescence. But applications of the model can also lead to outcomes relevant to certain unresolved issues. For example, it is not yet clear whether structural development typically consists of fundamental personality reorganizations or whether its course involves a series of minor transformations of trends established relatively early in life (Emmerich, 1966b). In the present scheme, there is a continuum of levels extending from the most "central" at the top of the hierarchy to the most "peripheral" at the bottom. If and when structural change occurs, its locus in the hierarchy indicates the centrality of change. Moreover, because centrality is conceptualized here as a continuum, the model makes explicit the distinct possibility that change will occur at a level that falls somewhere between the most central and most peripheral regions of the structure. It is also possible that there will be concomitant changes at more than one level, an outcome of particular relevance to the orthogenetic principle, which implies that any change worthy of designation as "fundamental" *will* be accompanied by a reorganization of the structure as a whole (Werner, 1957). In sum, a variety of forms of ipsative structural change are derivable by considering the highest level of change together with the extent to which such change

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is accompanied by reorganization at other regions of the structure.

When several periods of life are compared, it becomes feasible to detect *rates* of structural change and *changes* in such rates (accelerations and decelerations). The ipsative model calls for two distinct measures of rate, the first dealing with the number of attributes differentiated and the second with the number of levels of hierarchical organization. Little is known about accelerations and decelerations in rates of change of either type, and there is the intriguing possibility that these are related to "critical periods" in structural development. There also exists the possibility that the two kinds of change will interact with one another. For example, during periods of especially rapid differentiation, hierarchization may initially regress (Werner, 1957), and then there may be increasing hierarchization if and when the individual's personality resources (coping mechanisms, cognitive capacities, etc.) are mobilized to produce a higher level of integration. Or, conversely, when differentiation is proceeding slowly and personality resources are increasing rapidly, the emergence of transformed organizers could induce the acquisition of certain new attributes and the dropping out of others.

Person Sampling

Insofar as its inferences are limited to individuals, ipsative analysis is tied to a strictly idiographic approach to personality. However, there is no reason why ipsative structural analysis cannot take the second inferential step of seeking generalizations that apply to broader populations (Broverman, 1962). Thus, ipsative structural generalization would appear to involve a two-phase inferential process, the first phase consisting of generalization to individual lives, and the second of generalization to larger populations. Although methodologies for carrying out the first phase are relatively undeveloped, there is little reason to believe that insurmountable problems are involved. For example, the development of ipsative profiles of social motives has been studied by treating profile variance *within* age periods as "error" when determining the significance of profile variance (change) *between* age periods (Martin, 1964; Smock, 1966). Perhaps the greatest risk at this phase of inference is that replication will prove to be impossible. However, even this hazard can be reduced by means of multiple independent samplings within age periods (Martin, 1964).

The second and more familiar inferential step determines whether ipsative findings established for individuals can be generalized to broader populations of persons. This question of generalization can and perhaps should be raised for each of the noted properties of ipsative repertoires and structures. For example, it would not be surprising to find general age trends with respect to changing rates of differentiation and structural reorganization.

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Concept of Individuality

In a certain sense, the ipsative view is more thoroughly "person-oriented" than either the classical or differential approach (G. W. Allport, 1961, 1962). It goes beyond locating individuals on single or multiple dimensions and deliberately seeks generalizations (at the first phase of inference) that are highly specific and comprehensive accounts of individual lives. However, this is not to imply that ipsative repertoires or structures are necessarily unique. Indeed, the present analysis suggests that the issue of uniqueness has no special relevance for an ipsative approach. This point was made explicitly when the problem of attribute sampling was discussed. It was also implied when the general nature of ipsative structural properties was discussed and the importance of generalizing ipsative findings to broader populations was emphasized. Of course, it is always possible that the course of structural development is unique in certain persons, but this remains a matter for empirical demonstration.

IMPLICATIONS

Summarizing each view very briefly, the classical approach posits a series of levels of personality organization through which individuals pass in a fixed developmental order at differing rates; differential studies investigate consistency and change over time in the structure of individual differences on multiple attributes; and ipsative analysis considers the development of the individual's repertoire of attributes and their internal organization.

Despite these differences in focus, however, there are important similarities among the views. Clearly each seeks to find general principles of structural development while also providing information regarding individuality. They share a common interest in patterns of covariation as definers of personality structures, and all seem likely to benefit from systematic applications of hierarchical analysis. It is also apparent that the three approaches call ideally for multivariate longitudinal designs.

Moreover, all three views imply that structures can generate forces which, in turn, lead to structural change. Within the classical approach, for example, it has been hypothesized that developmental progressions are induced when the individual actively confronts that level on the developmental scale which is just above his characteristic level of functioning (Turiel, 1966). Because each individual cannot realize all human potentialities, certain differential accounts emphasize the necessity for successive personal choices among qualitatively different commitments which become increasingly refined in the course of development (Tyler, 1959, 1965). In this regard, differential theory might well draw upon an analogy between the genealogy of structural differentiations in personality development and

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processes of biological speciation (Mayr, 1963). In both cases, intrinsic selective forces (dispositional and genetic, respectively) can be seen as interacting with selective environmental factors to produce the "evolution" of structural forms. Within an ipsative frame, it has been suggested that structural imbalance (Heider, 1958) or cognitive dissonance (Festinger, 1957) can produce forces leading to structural change. In the present scheme, for example, incompatible elements at the higher levels of hierarchical structures might be expected to induce structural reorganization at a subsequent age period. Indeed, the presence of incompatible elements could provide an impetus for increasing hierarchization and for the differentiation of certain new attributes.

This common interest in the structural determinants of development suggests that the three views considered here go beyond approaches that give primary attention to processes of attribute learning, performance, and extinction (e.g., Bandura & Walters, 1963; Bijou & Baer, 1963). This is not to say that a theory must take a stand on whether it locates primary influences "within the person" or "in the environment." On the contrary, any satisfactory account of structural development will need to conceptualize both types of influences and will have much to say on the precise nature of their interrelations. Within the classical tradition, for example, it has been suggested that environmental factors serve primarily as accelerators or inhibitors of intrinsic developmental trends (Kohlberg, 1964, 1966; Wohlwill, unpublished MS). In any case, the essential point is that structural properties are not only the "consequences" of environmental influences, but that they also serve as determinants of response acquisitions and performance.

There are two ways that research efforts might achieve a better match between conceptual models and research procedures in this area. First, we are reminded of the importance of applying analytic procedures strictly in accordance with the conceptual requirements of each view. To illustrate, it will be recalled that although each approach derives structural patterns from an index of association or covariation, the type of index called for as well as the kind of information entering into it differs among the views. Second, there are certain historically based links between theory and method that could turn out to be essentially arbitrary. For example, only the ipsative approach traditionally has sought extensive samplings of the individual's behavior repertoire. But the first inferential step of ipsative analysis might also be applied in studies of developmental scale progressions, resulting in tests of classical hypotheses in the full context of individual lives.

We have seen how the same general concept can take on a different meaning in the context of each view. For example, "structural consistency over time" is signified in the classical approach by the absence of change, but can be manifested differentially by the presence of certain kinds of

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TABLE 1

EXAMPLES OF EVIDENCE WITHIN EACH APPROACH BEARING ON SEQUENTIALITY

Conclusion from Evidence	Evidence within Approach: Classical	Differential	Ipsative
Invariant sequential pattern	Few or no individual differences in order of progression on developmental scale.	Branchings of the structural genealogy are based on dimensions which emerge in a fixed order.	Few or no individual differences in sequencing of attribute differentiation and/or in sequences of processes underlying hierarchical integration.
Systematic variation in sequential patterning	Different sequential orders apply to various subpopulations and/or areas of personality functioning.	Different sequences of dimensional emergence apply to various branchings of the structural genealogy.	Similarity of attribute repertoires among individuals with different sequences of attribute differentiation in course of development.
Nonsequential patterning	Extensive individual and/or personality area differences in order of progression on developmental scale.	Same basic dimensions (continuity) and groupings of persons (stability) occur throughout course of development.	Extensive individual differences in nature of attributes differentiated and/or in sequences of processes underlying hierarchical integration.

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change. Perhaps the phenomena of structural development are too diverse to be ordered parsimoniously by one set of concepts. In any event, explicit recognition of differences in conceptual translation should lead to greater theoretical precision.

Although the three frameworks traditionally have emphasized different issues, each formulation has a broader scope than is generally recognized. For example, while only the classical view has been concerned explicitly with invariant sequences in structural development, this question does have relevance for the differential and ipsative approaches. To illustrate this point, Table 1 provides examples of evidence within each framework that would be appropriate for arriving at conclusions about sequential hypotheses.

Each of the nine possible outcomes listed in Table 1 (as well as others not listed) stands on its own as a potentially viable theory of structural development. This implication attests to the special importance of sequential analysis for research in this area. Moreover, derivations such as those found in Table 1 can help clarify the research implications of theories not discussed in the present review. A prominent example is Erikson's (1950, 1959) outline of psychosocial development, which, in present terms, hypothesizes an invariant sequential pattern within a differential framework. In Erikson's theory, the several psychosocial crises (trust versus mistrust, autonomy versus doubt, etc.) emerge in a fixed developmental order, and, in addition, the outcome of each crisis is conceptualized in bipolar terms, is seen as having a relatively permanent influence, and solutions to earlier-stage dilemmas are presumed to interact with later crises to produce differing outcomes. Empirical validation of this model would therefore demonstrate the presence of a genealogy of subpopulation differentiations based upon the presumed ordering of crises and bipolar outcomes.

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